Problem-Solution Fit

Date	10 July 2026
Team ID	xxxxxx
Project Name	Raising Waters
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS]

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Target Audience: Emergency Management Agencies, Local Disaster Coordinators, Municipal Flood Control Teams, and Regional Weather Evaluators.

6.
CUSTOMER LIMITATIONS
EG. BUDGET, DEVICES **[CL]**

Constraints: Tight local government operational budgets, limited hardware infrastructure (often relying on standard web browsers/laptops), and the need for zero-latency processing without relying on expensive computing clusters.

5.
AVAILABLE SOLUTIONS PROS & CONS **[AS]**

Manual Monitoring Tools:

Pros: Highly reliable local sensors.

Cons: Extremely slow, requires manual calculation, and lacks automated machine learning evaluation features.

Standard Weather Forecasts:

Pros: Good general coverage.

Cons: Not localized enough to provide distinct flood-tier threat levels for specific disaster boundaries.

2. PROBLEMS / PAINS + ITS FREQUENCY **[PR]**

Pains: High frequency of severe monsoon rain events causing rapid inundation.

Core Issue: Lack of instant, real-time warning data visibility makes it difficult to assess flood risk before it impacts residential sectors.

9. PROBLEM ROOT / CAUSE [RC]

Root Cause: Raw meteorological data is rarely integrated directly into automated predictive algorithms locally. This missing step stalls immediate interpretation, delaying the window required to issue safety alerts.

7. BEHAVIOR
+ ITS INTENSITY **[BE**

Current Actions: Coordinators intensely monitor multiple disjointed screens, manual spreadsheets, and external browser tabs simultaneously during peak monsoon emergencies to manually estimate threat levels.|

3. TRIGGERS TO ACT [TR]

Triggers: Escalating rainfall rates, automated severe weather alerts, sudden rises in local water levels, or high-risk predictive metrics flagged by incoming meteorological streams.

4. EMOTIONS BEFORE / AFTER **[EM]**

Before: Anxiety, stress, and feeling overwhelmed due to delayed alerts, fragmented data, and uncertain prediction timelines.
After: Confidence, clarity, and relief provided by a centralized visual dashboard showing immediate risk tiers and clear, real-time analytics.

10. YOUR SOLUTION [SL]

The Solution (Rising Waters): A dedicated, streamlined platform featuring automated Data Ingestion from live weather APIs, local ML Evaluation (via specialized .pkl models), and a clean Flask Dashboard that instantly updates risk tiers and archives predictions into logs.

8. CHANNELS OF BEHAVIOR [CH]

Online: Local network servers, centralized web dashboards, automated notification strings, and live API data streaming feeds.

OFFLINE

Offline: On-ground coordination centers, emergency radio channels, and physical control room stations.